

Kalman Filter

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Abstract

A thorough kalman filter derivation.

1 The System Model

A linear time-invariant discrete-time system governed by the state-space equations:

$$x_k = Ax_{k-1} + B_u u_{k-1} + w_{k-1} \quad (1)$$

$$y_k = Cx_k + v_k \quad (2)$$

where:

- $x_k \in \mathbb{R}^n$ is the state vector.
- $u_k \in \mathbb{R}^m$ is the control input vector.
- $y_k \in \mathbb{R}^p$ is the measured output.
- $w_k \in \mathbb{R}^n$ is the process noise with covariance $Q = E[w_k w_k^T]$.
- $v_k \in \mathbb{R}^p$ is the measurement noise with covariance matrix $R = E[v_k v_k^T]$, independent of w_k .

2 The Kalman Filter

The filter is defined as:

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + B_u u_{k-1} \quad (3)$$

The error dynamics associated is obtained as follows,

$$e_{k|k-1} = x_k - \hat{x}_{k|k-1} \quad (4)$$

$$= Ax_{k-1} + B_u u_{k-1} + w_{k-1} - A\hat{x}_{k-1|k-1} - B_u u_{k-1} \quad (5)$$

$$e_{k|k-1} = A e_{k-1|k-1} + w_{k-1} \quad (6)$$

The covariance matrix $P_{k|k-1}$ is given by $E[e_{k|k-1}e_{k|k-1}^T]$ as follows,

$$P_{k|k-1} = E[e_{k|k-1}e_{k|k-1}^T] \quad (7)$$

$$= E[(Ae_{k-1|k-1} + w_{k-1})(Ae_{k-1|k-1} + w_{k-1})^T] \quad (8)$$

$$= E[Ae_{k-1|k-1}e_{k-1|k-1}^T A^T] + E[Ae_{k-1|k-1}w_{k-1}^T] \quad (9)$$

$$+ E[w_{k-1}e_{k-1|k-1}^T] + E[w_{k-1}w_{k-1}^T] \quad (10)$$

$$= AE[e_{k-1|k-1}e_{k-1|k-1}^T]A^T + E[w_{k-1}w_{k-1}^T] \quad (11)$$

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q \quad (12)$$

The state estimation is corrected in a Luenberger fashion as follows,

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - C\hat{x}_{k|k-1}) \quad (13)$$

where the error is obtained as,

$$e_{k|k} = x_k - \hat{x}_{k|k} \quad (14)$$

$$= x_k - \hat{x}_{k|k-1} - K_k (y_k - C\hat{x}_{k|k-1}) \quad (15)$$

$$= x_k - \hat{x}_{k|k-1} - K_k (Cx_k + v_k - C\hat{x}_{k|k-1}) \quad (16)$$

$$= e_{k|k-1} - K_k C e_{k|k-1} - K_k v_k \quad (17)$$

$$e_{k|k} = (I - K_k C)e_{k|k-1} - K_k v_k \quad (18)$$

and its covariance is calculated as,

$$P_{k|k} = E[e_{k|k}e_{k|k}^T] \quad (19)$$

$$= E[(I - K_k C)e_{k|k-1} - K_k v_k](e_{k|k-1}(I - K_k C)^T - v_k^T K_k^T) \quad (20)$$

$$= E[(I - K_k C)e_{k|k-1}e_{k|k-1}^T(I - K_k C)^T] - E[(I - K_k C)e_{k|k-1}v_k^T K_k^T] \quad (21)$$

$$- E[K_k v_k e_{k|k-1}^T(I - K_k C)^T] + E[K_k v_k v_k^T K_k^T] \quad (22)$$

$$= (I - K_k C)E[e_{k|k-1}e_{k|k-1}^T](I - K_k C)^T - (I - K_k C)E[e_{k|k-1}v_k^T]K_k^T \quad (23)$$

$$- K_k E[v_k e_{k|k-1}^T](I - K_k C)^T + K_k E[v_k v_k^T]K_k^T \quad (24)$$

$$= (I - K_k C)E[e_{k|k-1}e_{k|k-1}^T](I - K_k C)^T + K_k E[v_k v_k^T]K_k^T \quad (25)$$

$$P_{k|k} = (I - K_k C)P_{k|k-1}(I - K_k C)^T + K_k R K_k^T \quad (26)$$

The covariance equation is expanded as follows,

$$P_{k|k} = (I - K_k C)P_{k|k-1}(I - K_k C)^T + K_k R K_k^T \quad (27)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - P_{k|k-1} C^T K_k^T - K_k C P_{k|k-1} C^T K_k^T + K_k R K_k^T \quad (28)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - P_{k|k-1} C^T K_k^T + K_k (C P_{k|k-1} C^T + R) K_k^T \quad (29)$$

This equation is used to choose the Kalman gain K_k with the following problem,

$$\min_{K_k} \text{Tr}(P_{k|k}) \quad (30)$$

by utilizing

$$\frac{\partial \text{Tr}(AB)}{\partial A} = B^T \quad (31)$$

and

$$\frac{\partial \text{Tr}(ACA^T)}{\partial A} = 2AC \quad (32)$$

The following is obtained,

$$\frac{\partial \text{Tr}(P_{k|k})}{\partial K_k} = -2(P_{k|k-1}C^T)^T + 2K_k(CP_{k|k-1}C^T + R) \quad (33)$$

$$= 0 \quad (34)$$

which gives,

$$2(P_{k|k-1}C^T) = 2K_k(CP_{k|k-1}C^T + R) \quad (35)$$

$$P_{k|k-1}C^T = K_k(CP_{k|k-1}C^T + R) \quad (36)$$

$$P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1} = K_k \quad (37)$$

$$K_k = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1} \quad (38)$$

$$K_k = P_{k|k-1}C^T S_k^{-1} \quad (39)$$

where $S_k = CP_{k|k-1}C^T + R$.

The covariance equation

$$P_{k|k} = (I - K_k C)P_{k|k-1}(I - K_k C)^T + K_k R K_k^T \quad (40)$$

using $K_k S_k = P_{k|k-1}C^T$ when examined gives,

$$P_{k|k} = (I - K_k C)P_{k|k-1}(I - K_k C)^T + K_k R K_k^T \quad (41)$$

$$= (P_{k|k-1} - K_k C P_{k|k-1})(I - C^T K_k^T) + K_k R K_k^T \quad (42)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - P_{k|k-1} C^T K_k^T + K_k C P_{k|k-1} C^T K_k^T + K_k R K_k^T \quad (43)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - P_{k|k-1} C^T K_k^T + K_k (C P_{k|k-1} C^T + R) K_k^T \quad (44)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - (K_k S_k) K_k^T + K_k S_k K_k^T \quad (45)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} - K_k S_k K_k^T + K_k S_k K_k^T \quad (46)$$

$$= P_{k|k-1} - K_k C P_{k|k-1} \quad (47)$$

$$P_{k|k} = (I - K_k C)P_{k|k-1} \quad (48)$$

3 Summary

With initial choices of $\hat{x}_0$, P_0 , the Kalman filter is summarized as follows:

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + B_u u_{k-1} \quad (49)$$

the filter is:

$$S_k = CP_{k|k-1}C^T + R \quad (50)$$

$$K_k = P_{k|k-1}C^TS_k^{-1} \quad (51)$$

then,

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - C\hat{x}_{k|k-1}) \quad (52)$$

$$P_{k|k} = (I - K_k C)P_{k|k-1} \quad (53)$$

4 Two-Stage Estimation Approach

In the presence of an unknown disturbance w_k , the discrete-time measurement equation is given by:

$$y_k = Cx_k + D_w w_k + v_k \quad (54)$$

To decouple the state x_k from the disturbance w_k during the initial estimation phase, a projection matrix Π is defined such that it nullifies the measurement matrix C . Assuming C has full column rank, Π is computed via the pseudo-inverse as:

$$\Pi = I - C(C^T C)^{-1}C^T \quad (55)$$

which satisfies $\Pi C = 0$.

4.1 Stage 1: Disturbance Estimation

Multiplying the measurement equation by the projection matrix yields the projected measurement $\bar{y}_k$:

$$\bar{y}_k = \Pi y_k \quad (56)$$

$$= \Pi C x_k + \Pi D_w w_k + \Pi v_k \quad (57)$$

$$= \Pi D_w w_k + \bar{v}_k \quad (58)$$

where $\bar{v}_k = \Pi v_k$ is the projected measurement noise with covariance $\bar{R} = \Pi R \Pi^T$. This projection completely removes the influence of the state x_k . Assuming the disturbance evolves as a random walk model ($w_k = w_{k-1} + w_{w,k-1}$ with process noise covariance Q_w), the Kalman filter for the disturbance is formulated as follows.

The prediction step for the disturbance is:

$$\hat{w}_{k|k-1} = \hat{w}_{k-1|k-1} \quad (59)$$

$$P_{k|k-1}^w = P_{k-1|k-1}^w + Q_w \quad (60)$$

The update step uses the projected measurement:

$$S_k^w = (\Pi D_w) P_{k|k-1}^w (\Pi D_w)^T + \bar{R} \quad (61)$$

$$K_k^w = P_{k|k-1}^w (\Pi D_w)^T (S_k^w)^{-1} \quad (62)$$

$$\hat{w}_{k|k} = \hat{w}_{k|k-1} + K_k^w (\Pi y_k - \Pi D_w \hat{w}_{k|k-1}) \quad (63)$$

$$P_{k|k}^w = (I - K_k^w \Pi D_w) P_{k|k-1}^w \quad (64)$$

4.2 Stage 2: State Estimation

With the updated disturbance estimate $\hat{w}_{k|k}$ acquired, it is treated as a known input for the primary state estimator. The state prediction relies on the prior disturbance estimate $\hat{w}_{k-1|k-1}$:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1} + B_{ud} u_{k-1} + B_{wd} \hat{w}_{k-1|k-1} \quad (65)$$

$$P_{k|k-1}^x = A_d P_{k-1|k-1}^x A_d^T + Q_x \quad (66)$$

For the state update, the effect of the newly estimated disturbance is subtracted from the actual measurement, creating a modified innovation term. The state update is executed as:

$$S_k^x = C P_{k|k-1}^x C^T + R \quad (67)$$

$$K_k^x = P_{k|k-1}^x C^T (S_k^x)^{-1} \quad (68)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k^x (y_k - C \hat{x}_{k|k-1} - D_w \hat{w}_{k|k}) \quad (69)$$

$$P_{k|k}^x = (I - K_k^x C) P_{k|k-1}^x \quad (70)$$

This sequential architecture ensures that the disturbance is estimated without interference from the system states, providing a clean feedforward term for the state estimator's correction phase.